

Prefab Setup Guide

This guide explains how to set up all the prefabs for the Space Shooter game in Unity.

Player Prefab

1. Create an empty GameObject, name it "Player"
2. Add components:
 - **SpriteRenderer**: Assign the player sprite (triangle or spaceship)
 - **Rigidbody2D**: Set Body Type to "Kinematic"
 - **BoxCollider2D** or **PolygonCollider2D**: Set "Is Trigger" to true
 - **PlayerController** script
 - **HealthSystem** script
 - **AudioSource**: For sound effects
3. Create child object "FirePoint" at position (0, 0.5, 0)
4. Set Tag to "Player"
5. Set Layer to "Player"
6. Configure PlayerController:
 - Move Speed: 8
 - Fire Rate: 0.2
 - Assign Bullet Prefab
 - Assign Fire Point transform
7. Configure HealthSystem:
 - Max Health: 5
8. Drag to Prefabs folder

Bullet Prefab (Player)

1. Create empty GameObject, name it "PlayerBullet"
2. Add components:
 - **SpriteRenderer**: Small yellow rectangle or circle
 - **Rigidbody2D**: Body Type "Kinematic"
 - **BoxCollider2D**: "Is Trigger" = true
 - **Bullet** script
3. Set Tag to "PlayerBullet"
4. Set Layer to "Bullet"
5. Configure Bullet:
 - Speed: 15
 - Damage: 1
 - Is Player Bullet: true
6. Drag to Prefabs folder

Enemy Bullet Prefab

1. Duplicate PlayerBullet, name it "EnemyBullet"
2. Change sprite color to red

3. Set Tag to "EnemyBullet"
4. Configure Bullet:
 - Is Player Bullet: false
5. Drag to Prefabs folder

Basic Enemy Prefab

1. Create empty GameObject, name it "EnemyBasic"
2. Add components:
 - **SpriteRenderer**: Red square or enemy sprite
 - **Rigidbody2D**: Body Type "Kinematic"
 - **BoxCollider2D**: "Is Trigger" = true
 - **Enemy** script
 - **AudioSource**
3. Set Tag to "Enemy"
4. Set Layer to "Enemy"
5. Configure Enemy:
 - Enemy Type: Basic
 - Movement Pattern: Straight
 - Shooting Pattern: Single
 - Health: 1
 - Score Value: 100
 - Move Speed: 3
 - Assign Enemy Bullet Prefab
 - Power Up Drop Chance: 0.1
6. Drag to Prefabs folder

Fast Enemy Prefab

1. Duplicate EnemyBasic, name it "EnemyFast"
2. Change sprite to smaller/different color (cyan)
3. Configure Enemy:
 - Enemy Type: Fast
 - Movement Pattern: Zigzag
 - Shooting Pattern: None
 - Health: 1
 - Score Value: 150
 - Move Speed: 6
4. Drag to Prefabs folder

Tank Enemy Prefab

1. Duplicate EnemyBasic, name it "EnemyTank"
2. Change sprite to larger/different color (purple)
3. Configure Enemy:
 - Enemy Type: Tank
 - Movement Pattern: Straight
 - Shooting Pattern: Spread

- Health: 5
 - Score Value: 300
 - Move Speed: 1.5
4. Drag to Prefabs folder

Boss Enemy Prefab

1. Duplicate EnemyBasic, name it "Boss"
2. Make sprite much larger
3. Configure Enemy:
 - Enemy Type: Boss
 - Movement Pattern: Circular
 - Shooting Pattern: Burst
 - Health: 20
 - Score Value: 1000
 - Move Speed: 1
 - Fire Rate: 2
4. Drag to Prefabs folder

Power-Up Prefabs

Weapon Upgrade

1. Create empty GameObject, name it "PowerUpWeapon"
2. Add components:
 - **SpriteRenderer**: Orange/yellow star or "W" sprite
 - **Rigidbody2D**: Body Type "Kinematic"
 - **CircleCollider2D**: "Is Trigger" = true
 - **PowerUp** script
3. Set Tag to "PowerUp"
4. Configure PowerUp:
 - Type: WeaponUpgrade
 - Move Speed: 2
5. Drag to Prefabs folder

Health Recovery

1. Duplicate PowerUpWeapon, name it "PowerUpHealth"
2. Change sprite to green cross or heart
3. Configure PowerUp:
 - Type: Health
 - Health Amount: 2
4. Drag to Prefabs folder

Shield

1. Duplicate PowerUpWeapon, name it "PowerUpShield"
2. Change sprite to blue circle or shield icon
3. Configure PowerUp:
 - Type: Shield
 - Shield Duration: 5

4. Drag to Prefabs folder

Background Prefab

1. Create empty GameObject, name it "ScrollingBackground"
2. Add child "Background1":
 - Add SpriteRenderer with space/star background
 - Set Sorting Layer to "Background"
 - Scale to fill screen
3. Add child "Background2":
 - Duplicate of Background1
 - Position above Background1
4. Add **ScrollingBackground** script to parent
5. Configure:
 - Scroll Speed: 2
 - Reset Height: 20 (based on sprite height)
6. Drag to Prefabs folder